

MTM4692-APPLIED SQL LIBRARY MANAGEMENT SYSTEM

GithubRepo: <https://github.com/rabiaz/SQL-Library-Project>

Drive link:

https://drive.google.com/drive/folders/1SfUBClol_Pf1QHiJaIFGIKejnRGW6GyE?usp=sharing

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Library Management System Database Design

1. Introduction

This project presents the design of a relational database system for managing multiple library branches in a centralized way. The system aims to improve how users search for books, track borrowing activity, and interact with library resources. Unlike traditional single-library systems, this project supports multiple libraries and integrates user activity, book availability, and review mechanisms into one unified structure.

2. Problem Statement

The real-world problem this project addresses is the lack of coordination and centralized data management across multiple library branches. In many library systems, information is fragmented. Each library branch operates independently, making it difficult for users to determine where a specific book is available. Borrowing records are often stored locally within each branch, preventing a unified view of user activity and book circulation. This leads to several practical issues. Additionally, the absence of a centralized borrowing system makes it difficult to track user behavior, such as how many books a user has borrowed or which books are most popular across all branches. By centralizing data about users, libraries, books, and borrowing transactions into a single relational database system, this project aims to eliminate these inefficiencies. The system ensures that users can access accurate book availability, track borrowing history, and rely on reviews that are only submitted by verified borrowers.

3. Why It Matters

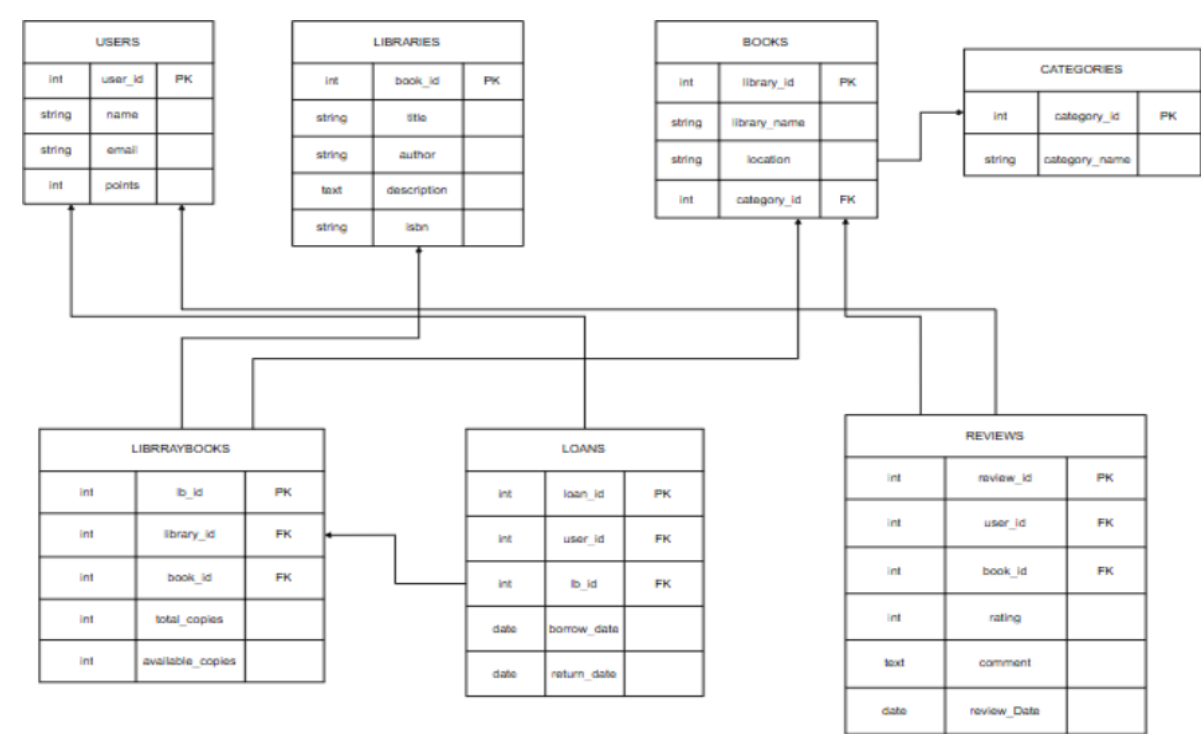
This project is important because it provides a structured and reliable solution to common inefficiencies in library systems. By centralizing data across multiple library branches, the system significantly improves access to books, allowing users to quickly identify where a specific title is available. In addition, it enables comprehensive tracking of borrowing history for each user, which supports both user experience and system management. A key feature of the system is its verified review mechanism, which ensures that only users who have actually borrowed a book can leave feedback. This increases the reliability and trustworthiness of reviews. Furthermore, the centralized database allows for meaningful data analysis, such as identifying popular books and active users, which can support better decision-making for library management.

4. Proposed System

The proposed system is designed to provide a centralized platform where users can interact with multiple libraries through a single database structure. Users will be able to search for books across different library branches and view their availability in real time. Once a book is located, users can borrow it from a specific library, and all borrowing activities will be recorded within the system. In addition, the system maintains a complete borrowing history for each user, enabling personalized tracking and activity analysis. A key component of the system is the review feature, which allows users to leave feedback only for books they have previously borrowed. This ensures that all reviews are based on real user experience, improving the overall quality and reliability of the information provided.

5. Preliminary Schema Design

Table Name	Primary Key	Foreign Keys	Purpose
Users	user_id	None	Stores user information such as name, email, and accumulated points.
Libraries	library_id	None	Stores information about different library branches and their locations.
Books	book_id	None	Contains general information about books such as title, author, and description.
LibraryBooks	library_book_id	library_id, book_id	Represents which books are available in which libraries and tracks the number of copies.
Loans	loan_id	user_id, library_book_id	Records borrowing transactions, including borrow date and return date.
Reviews	review_id	user_id, book_id	Stores user reviews and ratings for books. Only users who have borrowed a book are allowed to review it.



The relationships between the tables are carefully designed and strictly enforced to ensure data consistency and accurate representation of real-world interactions. The LibraryBooks table acts as a junction table between Books and Libraries, allowing the system to model a many-to-many relationship. This design enables a single book to exist in multiple library branches, while each library can store multiple books with different availability. The Loans table connects Users and LibraryBooks, making it possible to track which user borrowed which book from a specific library.

5. Proposed SQL Features

To make the system efficient, reliable, and capable of handling real-world library operations, several key SQL techniques are used in the design of this database:

- **Multi-Table Joins:** We use JOIN operations to combine data across multiple tables such as Users, Books, Libraries, and Loans. For example, JOINS allow the system to display which libraries contain a specific book or to show a user's borrowing history. LEFT JOINS can also be used to include users or books even if certain related data is missing, ensuring more complete query results.
- **Aggregation and Analysis:** Aggregate functions such as COUNT() and AVG() are used to analyze system data. For instance, COUNT() is used to calculate how many books a user has borrowed, while AVG() is used to determine the average rating of a book. These operations enable the system to generate meaningful insights like popular books and active users.
- **Database Views:** Views are used to simplify complex queries and provide user-friendly access to frequently needed data. For example, views can be created to display available books across libraries, summarize user borrowing activity, or present average ratings of books. This reduces the need to repeatedly write complex SQL queries.
- **Check Constraints:** CHECK constraints are used to enforce valid data entry. For example, rating values are restricted to a specific range (e.g., 1 to 5), and the number of available copies cannot exceed total copies. This helps maintain data accuracy and consistency.
- **Foreign Key Constraints:** Foreign keys are used to enforce relationships between tables and maintain referential integrity. For example, Loans are linked to both Users and LibraryBooks, ensuring that all borrowing records are valid. This prevents inconsistencies such as referencing non-existing users or books.
- **Business Rule Enforcement (Subqueries / Logic):** Additional logic is used to enforce important business rules. For example, a user is only allowed to submit a review for a book if they have previously borrowed it. This can be implemented using subqueries or conditional checks, ensuring that all reviews are based on real user experience.
- **Indexing (Optional Optimization):** Indexes can be applied to frequently searched fields such as book titles or user IDs to improve query performance, especially when searching for books or retrieving user activity.

6. Conclusion

This project presents a reliable and scalable database design for managing multiple library systems in a centralized way. By using a relational model, the system integrates book availability, borrowing operations, and user interactions into a single structured platform.

Compared to fragmented and isolated library systems, this approach provides a more efficient and consistent way to manage data across different branches. The use of SQL features such as joins, aggregation, and views enables accurate tracking of user activity and book circulation.

Additionally, the implementation of a verified review system ensures that feedback is based on actual user experience, increasing the overall reliability of the system. As a result, this database design not only improves accessibility and usability for users but also provides valuable insights for library management.